

The Access Residences — Blocks A1 & A2

Rooftop Photovoltaic System — Design, Supply, Installation & Commissioning

Prepared for: **Vasos Saouris**

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Ref: **LCY-PV-ACC-001**

Date: **September 2026**

38.94

kWp

INSTALLED PV
CAPACITY

170 m²

PANEL AREA (EPC
MET)

~71,300

kWh

EST. ANNUAL YIELD

~2.1 yrs

EST. PAYBACK
PERIOD

1. PROJECT UNDERSTANDING

Lighthief Cyprus Ltd is pleased to submit this technical and commercial proposal for the photovoltaic system at **The Access Residences**, Limassol — comprising **Blocks A1 and A2**.

The system is required to satisfy the project's **Energy Performance Certificate (EPC)**, which specifies a minimum installed PV capacity of **38.5 kWp**, based on a reference area of 255 m² at 15% module efficiency. Using modern high-efficiency N-Type modules ($\geq 22.9\%$ cell efficiency), the corresponding required panel area is **approximately 170 m²**.

EPC Compliance Achieved: Our proposed 66-panel array occupies exactly **170.0 m²** of roof area ($66 \times 2.28 \text{ m} \times 1.13 \text{ m}$), delivering **38.94 kWp** — meeting and marginally exceeding the 38.5 kWp EPC threshold.

The currently allocated roof area across both blocks is approximately **120 m²**. To reach the required 170 m², we propose utilising additional available roof surfaces including **staircase roof areas** and **roofs above enclosed roof-garden structures**, as indicated in the updated roof drawings. Our preliminary panel layout demonstrates how this can be achieved while maintaining adequate access and maintenance clearances.

2. PROPOSED SYSTEM

Block A1 — 19.47 kWp

33

PANELS

19.47

KWP

85 m²

PANEL AREA

- 33 × Jinko Tiger Neo N-Type 590 W panels
- 1 × Deye SUN-20K-SG01HP3 (20 kW hybrid)
- Installation across main flat roof + staircase roof
- 25° south-facing tilt, zero inter-row spacing
- DC/AC ratio: 0.97 — optimal for Cyprus irradiance

Block A2 — 19.47 kWp

33

PANELS

19.47

KWP

85 m²

PANEL AREA

- 33 × Jinko Tiger Neo N-Type 590 W panels
- 1 × Deye SUN-20K-SG01HP3 (20 kW hybrid)
- Installation across available flat roof + enclosed structure roofs
- 25° south-facing tilt, zero inter-row spacing
- DC/AC ratio: 0.97 — optimal for Cyprus irradiance

System Architecture

BLOCK A1 ROOF

BLOCK A2 ROOF

[33 × Jinko 590W panels]
Portrait, 25° tilt, south-facing
+ staircase roof area

[33 × Jinko 590W panels]
Portrait, 25° tilt, south-facing
+ enclosed structure roofs

| DC strings
| (4mm² solar cable)



Deye SUN-20K
SG01HP3
20 kW Hybrid
IP65 / HV-ESS
Battery-Ready

| AC 3-phase 400V



| DC strings
| (4mm² solar cable)



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EAC METERING & MAIN LV DISTRIBUTION
(Block A1 + Block A2 supply boards)
Net Billing — EAC / RAAEK connection applied

Monitoring: Deye SolarmanPro cloud platform (per block)
Battery Storage: Future upgrade via Deye HV ESS (inverters battery-ready)

Battery-Ready Architecture: Both Deye SUN-20K-SG01HP3 inverters support high-voltage battery storage (150–850 V HV ESS). Battery units can be retrofitted at any time without inverter replacement — this future-proofs the installation for potential storage requirements.

3. KEY EQUIPMENT SPECIFICATIONS

Jinko Tiger Neo N-Type 590 W

PV MODULE — N-TYPE TOPCON

Technology	N-Type TOPCon Bifacial
Power (STC)	590 Wp
Module Efficiency	22.9%
Dimensions	2278 × 1134 × 30 mm
Imp / Isc	13.36 A / 14.07 A
Vmp / Voc	44.17 V / 52.90 V
Product Warranty	12 years
Performance Warranty	30 years linear

Deye SUN-20K-SG01HP3

3-PHASE HYBRID INVERTER 20 KW

Rated AC Output	20 kW / 3-phase 400 V
Max DC Input	24 kW (DC oversizing OK)
MPPT Voltage Range	150 – 850 V
Number of MPPTs	2 independent MPPTs
Max PV Input Current	20 A per MPPT
Battery Type	HV ESS (150–850 V)
Functions	Hybrid / Off-Grid / Asymmetric
Protection / Comms	IP65 / WLAN-FE / Load Meter
Product Warranty	10 years

Helios Aluminum Flat-Roof Mounting System

PORTRAIT ORIENTATION — 25° TILT, SOUTH-FACING (180°), ZERO INTER-ROW SPACING (3 ROWS × 1 PANEL DEEP PER BLOCK). BACK HEIGHT PER ROW: 0.96M.

Rail system	Helios T-bolt aluminum rail
Triangles/supports	Helios 1P, 5°–28° adjustable
Clamps	Helios mid + end clamps, 35 mm
Material	Anodised aluminium, stainless steel fixings
Roof type	Flat roof (concrete / screed)
Load bearing	Designed to EN 1991 wind/snow

DC / AC Protection & BOS

BALANCE OF SYSTEM — PER IEC 60364 & BS 7671

DC solar cable	4 mm² halogen-free, 1500 V DC
MC4 connectors	IP67, 30 A rated
DC MCBs	Helios DC 500 V, per string
DC surge protection	Type II SPD, per inverter
AC protection	RCBO Type-A, AC SPD Type II
Smart meters	3-phase CT energy meter × 2
Conduit / trays	Metal tracking, IP-rated boxes
Earthing	Earth rods, bonding per IEC 60364

4. PRELIMINARY PANEL LAYOUT

The following layout is based on the architectural roof drawings provided. Panel rows run **E–W** (perpendicular to the building's N25°E long axis), with panels tilted at 25° facing due south. The configuration uses **3 rows × 1 panel deep** per row in each block, with zero inter-row spacing.

LOCATION	DESCRIPTION	PANELS	AREA (M²)	CAPACITY (KWP)
Block A2 — Main flat roof (NE portion)	11 panels wide (E–W) × 3 rows deep (N–S), south-facing, 25° tilt. Positioned on the upper flat section of Block A2 as indicated in the architectural drawing, clear of staircase and MEP equipment.	33	77	19.47
Block A1 — Main flat roof (NW portion)	11 panels wide (E–W) × 3 rows deep (N–S), south-facing, 25° tilt. Positioned in the upper flat section of Block A1 as indicated in the architectural drawing, clear of the staircase enclosure and roof access.	33	77	19.47
Total — Blocks A1 + A2 Combined		66	154 m²	38.94 kWp

Layout method — 3 rows × 1 panel deep @ 25° tilt, zero inter-row spacing: Each of the 3 rows is a single panel deep (2.28m inclined), tilted at 25° facing south. Back height per row = 0.96m (within the 1.20m architectural constraint). Shadow from the top of each row lands exactly at the base of the next row at 10am on 21 December — the mathematical zero-gap condition. No inter-row spacing required. Total N–S footprint per block: $3 \times (2.28 \times \cos 25^\circ) = \mathbf{6.19m}$. Total E–W footprint: $11 \times 1.13m = \mathbf{12.43m}$. Total footprint per block: **77 m²** — comfortably within the main flat roof areas indicated in the architectural drawing.

Confirmed on architectural drawing: Panel array positions match the zones allocated for PV on the roof drawings provided by the Architect (Roof Block A1 and Roof Block A2). Final exact positions, row offsets from parapets and MEP clearances to be confirmed at site survey. A panel layout drawing to be issued by Lighthief Cyprus's ETEK Electrical Engineer (Costas Hadjikyriacou) as part of the EAC application.

5. PRICE SCHEDULE (EXCL. VAT)

Equipment Included SUPPLY OF ALL MATERIALS • 66 × Jinko Tiger Neo N-Type 590 W bifacial PV modules (38.94 kWp) • 2 × Deye SUN-20K-SG01HP3 3-phase hybrid inverter, 20 kW (1 per block) • 2 × 3-phase smart energy meters with CT clamp sets • Aluminum flat-roof mounting system — adjustable tilt triangles, rails, clamps, SS hardware for 66 panels • DC solar cabling, MC4 connectors, string protection (DC MCBs, SPD Type II) • AC protection boards (RCBO, SPD Type II per block), cable trays, conduit, junction boxes, earth rods, labelling	Services Included TURNKEY DESIGN, INSTALLATION & PERMITTING • Electrical system design — SLD, string layout, protection coordination (ETEK licensed engineer) • EAC / RAAEK net billing application — Block A1 (Parts A + B + inspection) • EAC / RAAEK net billing application — Block A2 (Parts A + B + inspection) • Supply, delivery to site and full installation of all equipment • Commissioning, testing, CT verification, zero-export setup • Deye SolarmanPro cloud monitoring platform setup (both blocks) • Project documentation, warranty registration & handover package
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Equipment supply (all panels, inverters, mounting & BOS)	18,970.00
Design, installation, commissioning & permitting	8,100.00
Total excl. VAT	27,070.00
VAT @ 19%	5,143.30
TOTAL (Incl. VAT)	€ 32,213.30

Turnkey price per kWp (excl. VAT): €27,070 / 38.94 kWp = €695/kWp — fully installed, designed, permitted and commissioned including dual EAC grid connection for both blocks.

6. ENERGY YIELD & FINANCIAL SUMMARY

PARAMETER	VALUE	BASIS
Installed capacity	38.94 kWp	66 × Jinko 590 W STC
Estimated specific yield (Limassol, 25° south-facing, monofacial, PR 0.80)	1,712 kWh/kWp/yr	PVGIS-SARAH3 database, ERA5 meteo, 2005–2023 average, 14% system losses
Bifacial rear-side gain — white flat roof, albedo 0.45, bifaciality 80%	+7% ≈ +120 kWh/kWp/yr	Conservative estimate; Jinko BDV bifaciality factor 80±5% per datasheet
Effective specific yield (monofacial + bifacial)	1,832 kWh/kWp/yr	25° south-facing, bifacial on white membrane roof
Estimated annual generation	~71,300 kWh/yr	38.94 kWp × 1,832 kWh/kWp
Self-consumption (common areas — lifts, lighting, pumps, EV charging)	~49,900 kWh/yr	70% self-use assumption — to be confirmed against building load profile
Net billing export to EAC grid	~21,400 kWh/yr	30% monthly-period surplus at wholesale rate
Annual savings — self-consumed @ €0.22/kWh	~€10,985/yr	Avoided EAC electricity purchase
Annual revenue — export @ €0.095/kWh (RAAEK net billing rate)	~€2,033/yr	EAC net billing monetary credit on net surplus
Total annual financial benefit		~€13,018/yr
Simple payback period (on total excl. VAT)	~2.1 years	€27,070 / €13,018
25-year lifetime generation	~1,700 MWh	0.4% annual degradation (Jinko N-type linear warranty)

Yield based on PVGIS-SARAH3 irradiance data for Limassol (34.68°N, 33.04°E), 25° south-facing tilt as per architectural drawings, Jinko Tiger Neo N-Type 590 W modules, Deye SUN-20K-SG01HP3 inverters, performance ratio 0.80. Financial figures exclude VAT. Actual self-consumption depends on the building's common-area load profile; battery storage would increase self-consumption and reduce payback further.

7. SCOPE — INCLUSIONS & EXCLUSIONS

✓ Included in This Quotation

FULL TURNKEY SCOPE

- Supply of all PV panels, inverters, mounting, cabling and protection equipment
- Roof survey and final panel layout design
- Electrical system design (SLD) by ETEK licensed engineer
- All installation labour — panels, inverters, wiring, earthing
- EAC / RAAEK Net Billing application for both blocks (Parts A + B + inspection)
- Commissioning, testing and yield verification
- Deye SolarmanPro monitoring platform setup
- Project documentation and system handover
- Equipment datasheets and warranty certificates
- 2-year Lighthief installation workmanship warranty

✗ Not Included / Exclusions

TO BE SEPARATELY AGREED OR PROVIDED BY OTHERS

- VAT (19% applicable on all sums above)
- Structural assessment / engineer's certificate for roof loading (by structural engineer if required)
- Battery storage units (inverters are battery-ready; batteries are optional future addition)
- Internal consumer unit / distribution board modifications within apartments
- Metering arrangement between common areas and individual units (to be coordinated with project consultants per client's instruction)
- Crane hire or scaffold if required beyond standard access equipment
- Any civil or waterproofing works on roof
- EAC smart meter upgrade (if required by EAC at connection)

8. WARRANTIES

30

YEAR LINEAR PERFORMANCE

Jinko Tiger Neo panels — $\geq 87.4\%$ power output guaranteed at year 25, linear degradation schedule

12

YEAR PRODUCT WARRANTY

Jinko Tiger Neo panels — materials and workmanship; Deye inverters: 10 years product warranty

10

YEAR INVERTER WARRANTY

Deye SUN-20K-SG01HP3 — full manufacturer's product warranty, both units

2

YEAR INSTALLATION WARRANTY

Lighthief Cyprus — workmanship warranty on all installation work, cabling and mounting

25

YEAR MOUNTING WARRANTY

Helios aluminum mounting system — structural integrity of aluminium rails and support frames

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REMOTE MONITORING

Deye SolarmanPro — lifetime cloud monitoring platform included with inverter purchase

9. EAC NET BILLING APPLICATION & DELIVERY TIMELINE



EAC Application Timeline Note: EAC Part A (conditions of connection) typically takes 3–6 weeks; Part B and inspection a further 4–8 weeks. Total timeline from contract to grid-connected operation is estimated at **10–14 weeks**, subject to EAC processing times which are outside Lighthief's control. Installation can proceed ahead of grid connection, with the system energised on EAC confirmation.

10. COMMERCIAL TERMS

PAYMENT SCHEDULE

- **50%** — with signed contract & order placement (€13,535.00)
- **40%** — upon delivery of equipment to site (€10,828.00)
- **10%** — upon commissioning and system handover (€2,707.00)
- All amounts **exclude VAT** at 19%

PRICING & VALIDITY

- Prices valid for **30 days** from date of issue
- Equipment prices subject to confirmation at order placement
- Applicable currency: **Euro (€)**

DELIVERY & INSTALLATION

- Equipment delivery: approximately **2–3 weeks** from order
- Installation duration: approximately **3–4 working days** on site
- Client to ensure clear roof access and parking for installation team
- Site coordination with main contractor / architect where required

GENERAL CONDITIONS

- Binding offer subject to written contract and equipment stock confirmation
- Final metering/billing arrangement between common areas and individual units to be coordinated with project consultants — not included in this scope
- Structural roof assessment (if required) is client's responsibility
- Lighthief Cyprus Ltd — HE 477423

11. NEXT STEPS

To proceed, please confirm your acceptance of this proposal in writing. We will then issue a formal contract for signature and coordinate a brief site visit to finalise the panel layout before procurement.

For any questions regarding this proposal, please contact our Cyprus Director directly:

Alexander Papacosta — Cyprus Director, Lighthief Cyprus Ltd
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DOCUMENT

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ELECTRICAL ENGINEER

Costas Hadjikyriacou
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Confidential — prepared exclusively for Vasos Saouris / The Access Residences. Binding offer subject to written contract, equipment availability confirmation and final engineering sign-off. Energy yield estimates are indicative only and do not constitute a guaranteed performance undertaking.
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